

APPENDIX

A. Feasible configurations of the two-dimensional DSSR of EVCS

A.1 Three classes of hyperplanes

The two-dimensional DSSR of each EVCS, denoted as Ω^{DSSR} , is configured by a set of hyperplanes $a_k P + b_k Q \leq C_k$. Based on the signs of coefficients (a_k, b_k, C_k) , only four distinct combinations exist: $+++$, $++-$, $--+$ and $---$.

For clarity, hyperplanes with positive a_k, b_k are grouped into the set $\Omega^{\text{DSSR},+}$ (referred to as “Quadrant I-oriented”), while those with negative a_k, b_k form $\Omega^{\text{DSSR},-}$ (referred to as “Quadrant III-oriented”). Clearly, $\Omega^{\text{DSSR},+} \cup \Omega^{\text{DSSR},-} = \Omega^{\text{DSSR}}$. A positive C_k indicates the hyperplane lies in its expected quadrant without “jumping across” the coordinate origin $\mathbf{O}$; conversely, a negative C_k implies it “jumps across” $\mathbf{O}$, as illustrated in Fig. 1.

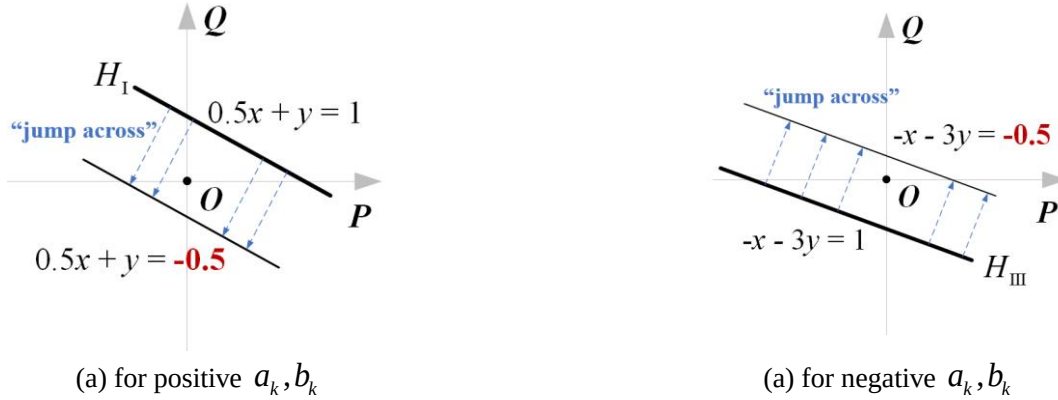


Fig. 1 Schematic diagrams: illustration of hyperplanes crossing/non-crossing the origin $\mathbf{O}$

Furthermore, hyperplanes in Ω^{DSSR} are classified into three types, as shown in Fig. 2:

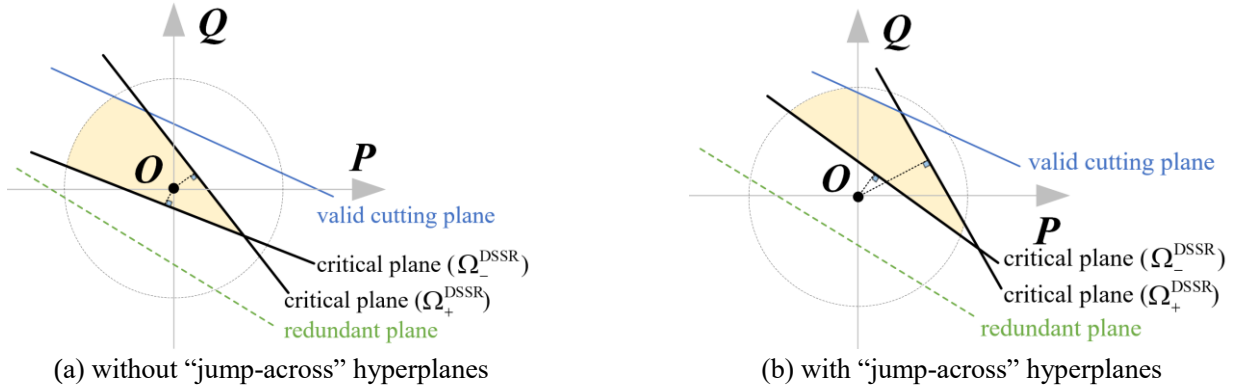


Fig. 2 Schematic diagrams: the three classes of hyperplanes

1) *critical planes*: An Ω^{DSSR} contains only two critical planes, one selected from $\Omega^{\text{DSSR},+}$ and the other from $\Omega^{\text{DSSR},-}$. Taking $\Omega^{\text{DSSR},+}$ as an example, its critical plane is defined as:

$$(a^{\text{critical}}, b^{\text{critical}}, C^{\text{critical}}) = \begin{cases} \arg \max_{a,b,C} d(a,b,C), & \text{if } (a,b,C) \in \{(a,b,C) \mid \Omega^{\text{DSSR},+}, C < 0\} \neq \emptyset \\ \arg \min_{a,b,C} d(a,b,C), & \text{otherwise} \end{cases} \quad (1)$$

where $d(a,b,C)$ denotes the perpendicular distance from the hyperplane $aP + bQ \leq C$ to the origin $\mathbf{O}$. This definition selects the farthest hyperplanes from $\mathbf{O}$ among those “jumping across” $\mathbf{O}$, and the closest otherwise.

2) *valid cutting planes*: Any hyperplanes that form the ultimate boundary of Ω^{DSSR} is classified as a valid cutting plane. By definition, critical planes are always valid cutting planes, so Ω^{DSSR} contains at least two such planes.

3) *redundant planes*: Hyperplanes that don't contribute to forming the boundary of Ω^{DSSR} are classified as redundant. Typically, Ω^{DSSR} contains a large number of redundant planes.

A.2 Analysis framework based on four categories of critical plane combinations

To systematically handle all possible configurations of Ω^{DSSR} in practical applications, we adopt a structured analysis framework. The methodology first classifies Ω^{DSSR} into one of the four categories based on the positions of its two critical planes, then analyzes the influence of remaining valid cutting planes.

The four categories of critical plane combinations are illustrated in Fig. 3:

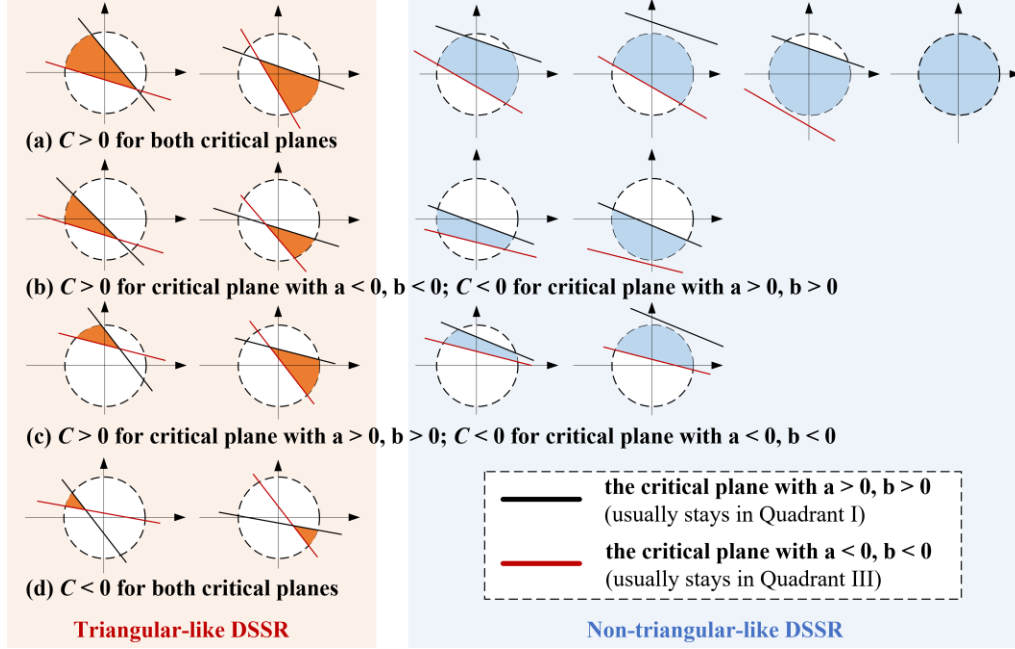


Fig. 3 Four categories of DSSR in terms of critical plane positions

The above framework is applied when calculating the security capacity of a given operating point (detailed in Section III.B), or determining the reference operating point during minute-level corrections, which ensures comprehensive and efficient analysis.

B. Supplement simulation results of other EVCSs

In the simulation part of this paper, EVCS3 and EVCS1 are selected as the representative EVCS for IEEE-33 and IEEE-141 bus test systems. Similar patterns also hold for other EVCSs, whose simulation results are supplemented as follows.

B.1 Supplement materials of IEEE-33 bus system

For EVCS1 and EVCS2, their respective security capacity comparisons between M1 and M2 over the whole day / morning peak / evening peak are displayed in Fig. 4 and Fig. 5.

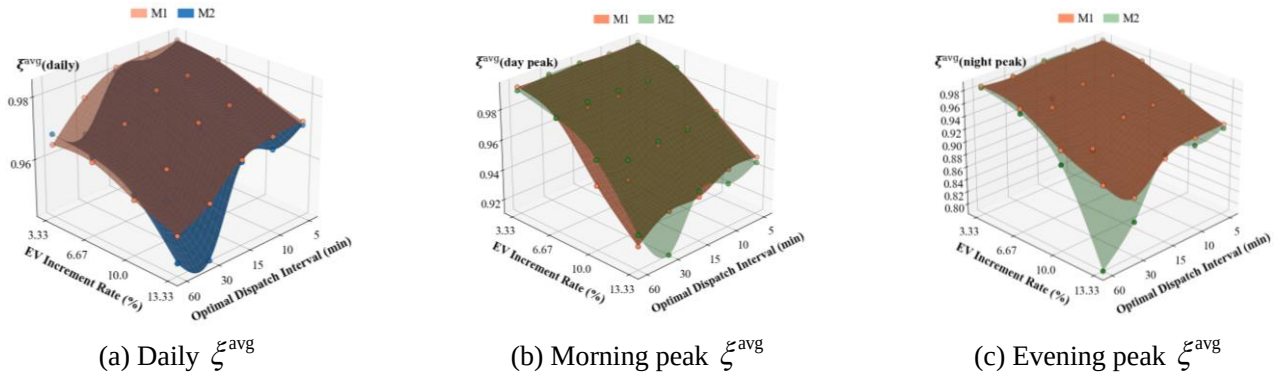


Fig. 4 Comparisons of M1 and M2 over daily/morning peak/evening peak average security capacity for EVCS1

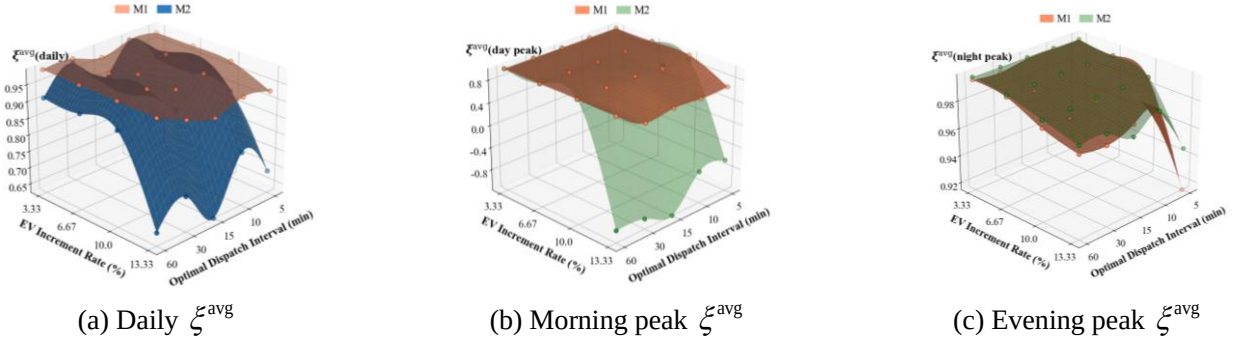


Fig. 5 Comparisons of M1 and M2 over daily/morning peak/evening peak average security capacity for EVCS2

The cumulative times of negative real-time security capacity of EVCS1 and EVCS2 under M2 are displayed in Fig. 6.

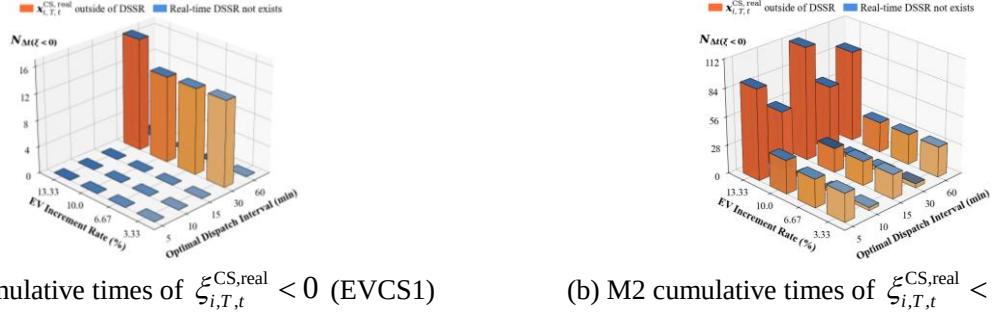


Fig. 6 M2 cumulative times of negative real-time security capacity for EVCS1 and EVCS2

B.2 Supplement materials of IEEE-141 bus system

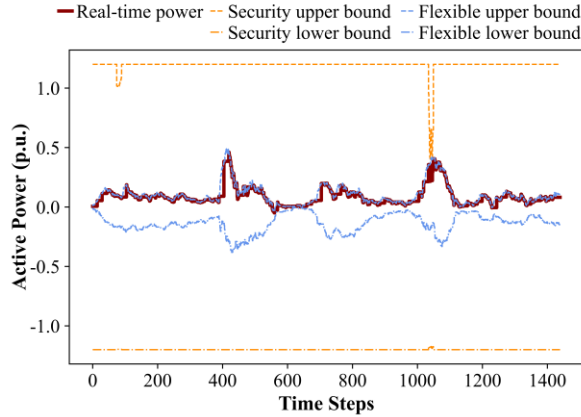
The comparative results between M1 and M2 of EVCS2~6 in the most adopted 15-minute optimal dispatch interval are displayed in TABLE 1.

TABLE 1
Economic and Security Performance of M1 and M2 under Different EV Increment Rates ($\Delta T=15$ min, 141-bus system)

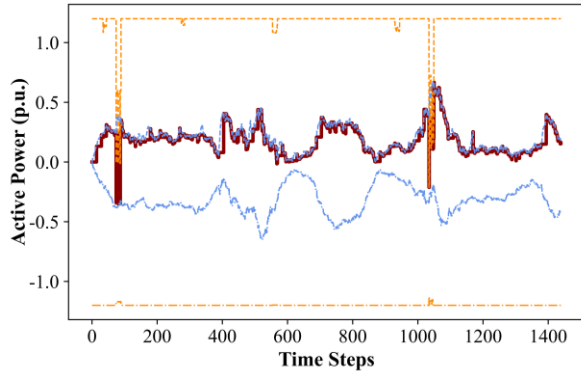
EVCS 2										
	$\gamma = 3.33\%$		$\gamma = 6.67\%$		$\gamma = 10.00\%$		$\gamma = 13.33\%$		$\gamma = 16.67\%$	
	M1	M2	M1	M2	M1	M2	M1	M2	M1	M2
$x_{i,T,t}^{\text{CS,real}}$ violation times (min)	0	0	0	0	0	0	0	0	0	2
ξ^{avg} (morning peak)	99.89%	99.90%	99.68%	99.69%	99.31%	99.32%	98.26%	98.35%	97.64%	97.81%
ξ^{avg} (evening peak)	99.68%	99.72%	99.32%	99.35%	98.60%	98.69%	96.75%	96.30%	95.04%	91.26%
Charging profit (\$)	732.70	734.66	803.21	805.16	878.08	880.04	975.90	977.86	1056.13	1058.08
Penalty cost (\$)	0	0	0	0	0	0	0	0	0	0
EVCS 3										
	$\gamma = 3.33\%$		$\gamma = 6.67\%$		$\gamma = 10.00\%$		$\gamma = 13.33\%$		$\gamma = 16.67\%$	
	M1	M2	M1	M2	M1	M2	M1	M2	M1	M2
$x_{i,T,t}^{\text{CS,real}}$ violation times (min)	0	15	0	26	0	30	0	30	0	30
ξ^{avg} (morning peak)	99.73%	99.74%	99.08%	99.14%	98.07%	98.32%	97.34%	97.55%	95.96%	96.20%
ξ^{avg} (evening peak)	99.11%	99.22%	96.64%	58.71%	90.23%	74.30%	89.77%	71.92%	87.13%	69.07%
Charging profit (\$)	1626.91	1630.07	1760.33	1763.44	1888.20	1891.31	2001.17	2004.26	2121.42	2124.41
Penalty cost (\$)	0	0	0	0	0	0	0	0	0	0
EVCS 4										
	$\gamma = 3.33\%$		$\gamma = 6.67\%$		$\gamma = 10.00\%$		$\gamma = 13.33\%$		$\gamma = 16.67\%$	
	M1	M2	M1	M2	M1	M2	M1	M2	M1	M2
$x_{i,T,t}^{\text{CS,real}}$ violation times (min)	0	0	0	0	0	0	0	0	0	15
ξ^{avg} (morning peak)	99.81%	99.79%	99.14%	99.22%	97.99%	98.23%	96.91%	97.17%	94.55%	95.13%

ξ^{avg} (evening peak)	99.33%	99.45%	98.41%	98.61%	95.68%	93.38%	91.35%	52.53%	89.43%	63.55%
Charging profit (\$)	1130.23	1132.87	1263.77	1266.42	1392.25	1394.80	1506.54	1509.16	1653.74	1655.97
Penalty cost (\$)	0	0	0	0	0	0	0	0	0	0
EVCS 5										
	$\gamma = 3.33\%$		$\gamma = 6.67\%$		$\gamma = 10.00\%$		$\gamma = 13.33\%$		$\gamma = 16.67\%$	
	M1	M2	M1	M2	M1	M2	M1	M2	M1	M2
$x_{i,T,t}^{\text{CS,real}}$ violation times (min)	0	0	0	0	0	0	0	0	0	0
ξ^{avg} (morning peak)	99.90%	99.91%	99.66%	99.69%	99.27%	99.31%	98.94%	98.95%	98.15%	98.20%
ξ^{avg} (evening peak)	99.62%	99.70%	98.82%	98.96%	98.19%	98.31%	97.46%	97.61%	96.19%	96.33%
Charging profit (\$)	609.57	610.21	677.60	678.07	777.84	778.46	848.66	849.23	953.79	954.43
Penalty cost (\$)	0	0	0	0	0	0	0	0	0	0
EVCS 6										
	$\gamma = 3.33\%$		$\gamma = 6.67\%$		$\gamma = 10.00\%$		$\gamma = 13.33\%$		$\gamma = 16.67\%$	
	M1	M2	M1	M2	M1	M2	M1	M2	M1	M2
$x_{i,T,t}^{\text{CS,real}}$ violation times (min)	0	0	0	0	0	8	0	15	0	15
ξ^{avg} (morning peak)	99.59%	99.63%	98.86%	98.94%	97.88%	98.05%	97.11%	97.21%	95.66%	95.82%
ξ^{avg} (evening peak)	99.43%	99.46%	97.86%	97.40%	95.96%	64.48%	89.81%	73.42%	86.49%	69.29%
Charging profit (\$)	1161.02	1164.03	1295.24	1298.19	1421.04	1424.00	1548.35	1551.29	1668.81	1672.04
Penalty cost (\$)	0	0	0	0	0	0	0	0	0	0

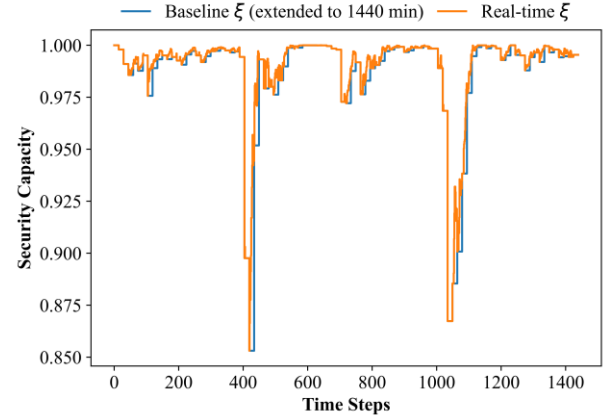
The power profiles and security capacity profiles of EVCS2~EVCS6 are presented in Fig. 7.



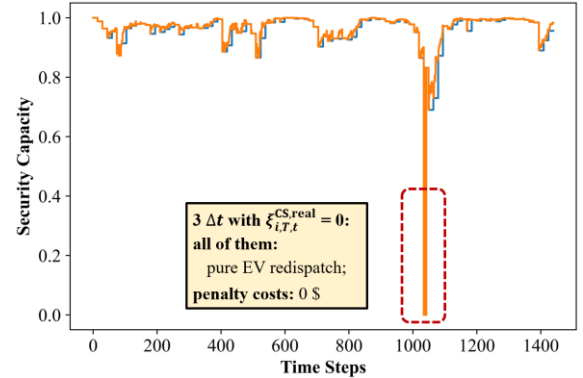
(a) Power profiles of EVCS2



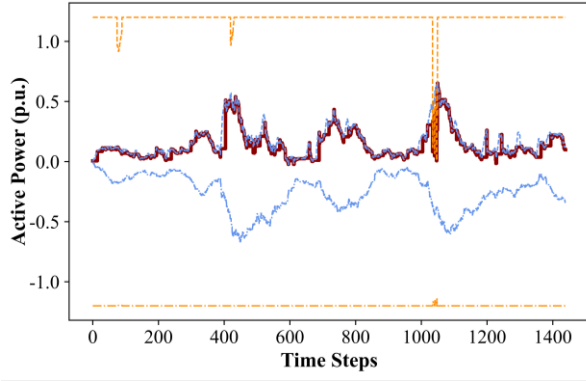
(c) Power profiles of EVCS3



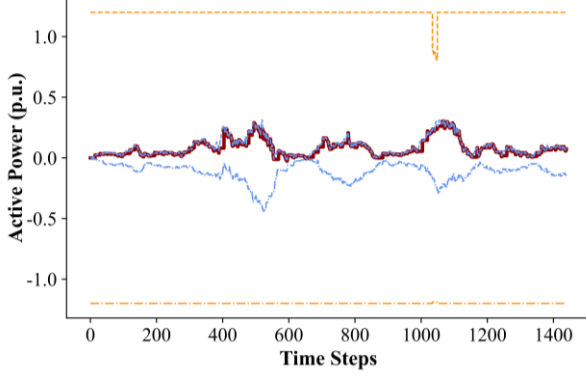
(b) Baseline and real-time security capacity of EVCS2



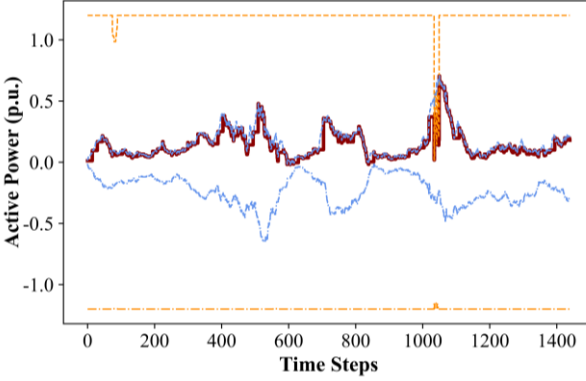
(d) Baseline and real-time security capacity of EVCS3



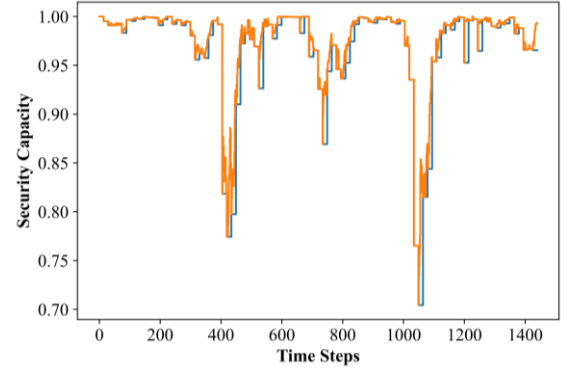
(e) Power profiles of EVCS4



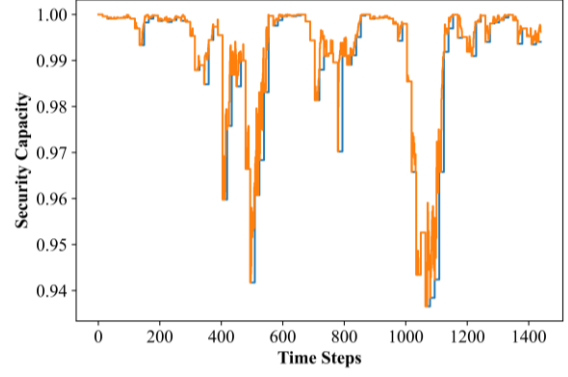
(g) Power profiles of EVCS5



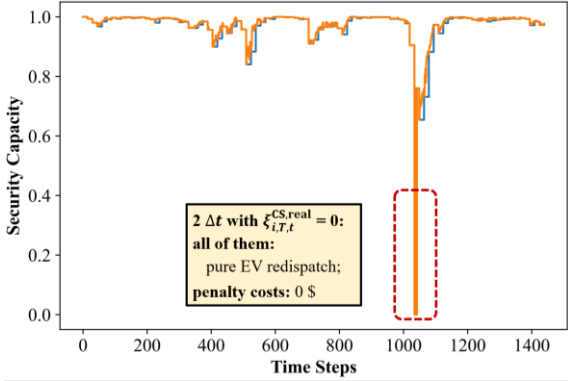
(i) Power profiles of EVCS6



(f) Baseline and real-time security capacity of EVCS4



(h) Baseline and real-time security capacity of EVCS5



(j) Baseline and real-time security capacity of EVCS6

Fig. 7 Operating point, security bounds and flexible bounds variations of EVCS2~6 (141-bus system)

For EVCS2~EVCS6, their respective security capacity comparisons between M1 and M2 over the whole day / morning peak / evening peak are displayed in Fig. 8 ~ Fig. 12.

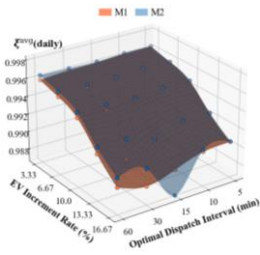
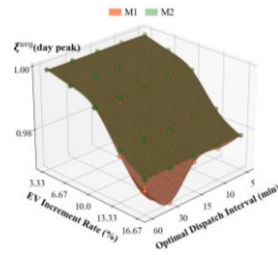
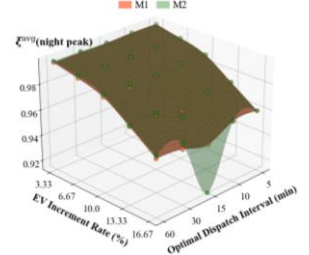
(a) Daily ξ^{avg} (b) Morning peak ξ^{avg} (c) Evening peak ξ^{avg}

Fig. 8 Comparisons of M1 and M2 over daily/morning peak/evening peak average security capacity for EVCS2 (141-bus)

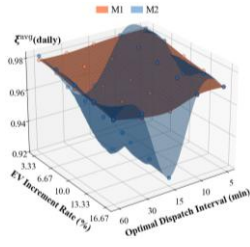
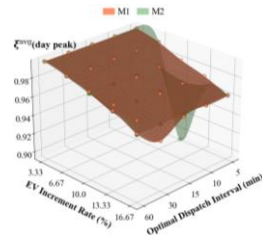
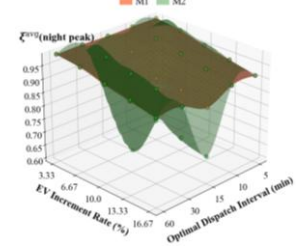
(a) Daily ξ^{avg} (b) Morning peak ξ^{avg} (c) Evening peak ξ^{avg}

Fig. 9 Comparisons of M1 and M2 over daily/morning peak/evening peak average security capacity for EVCS3 (141-bus)

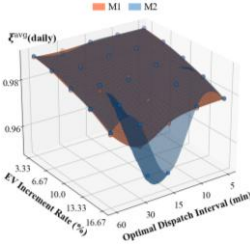
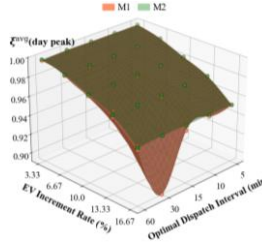
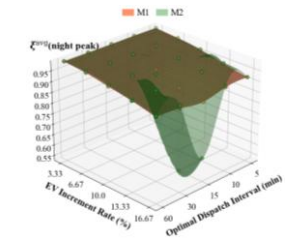
(a) Daily ξ^{avg} (b) Morning peak ξ^{avg} (c) Evening peak ξ^{avg}

Fig. 10 Comparisons of M1 and M2 over daily/morning peak/evening peak average security capacity for EVCS4 (141-bus)

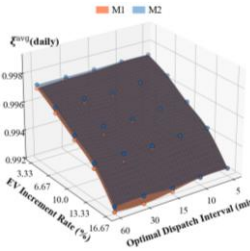
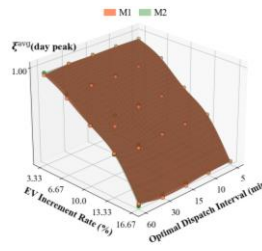
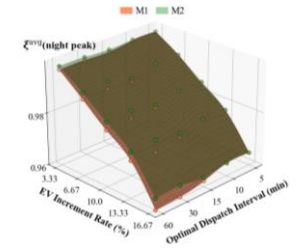
(a) Daily ξ^{avg} (b) Morning peak ξ^{avg} (c) Evening peak ξ^{avg}

Fig. 11 Comparisons of M1 and M2 over daily/morning peak/evening peak average security capacity for EVCS5 (141-bus)

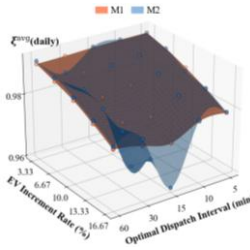
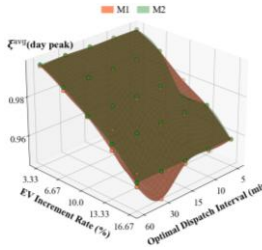
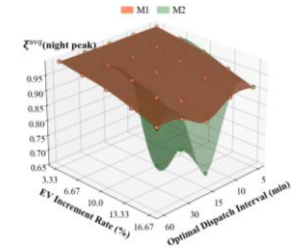
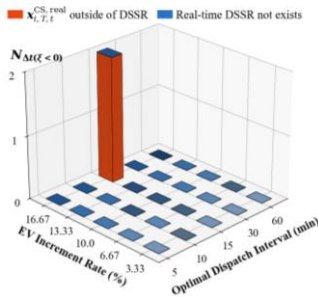
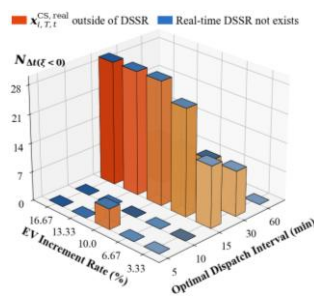
(a) Daily ξ^{avg} (b) Morning peak ξ^{avg} (c) Evening peak ξ^{avg}

Fig. 12 Comparisons of M1 and M2 over daily/morning peak/evening peak average security capacity for EVCS6 (141-bus)

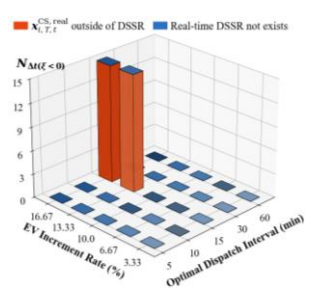
The cumulative times of negative real-time security capacity of EVCS2 ~ EVCS6 (based on M2) are displayed in Fig. 13.



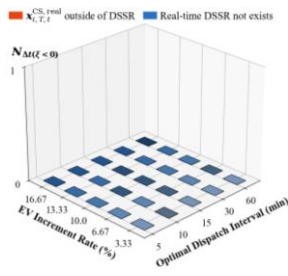
(a) EVCS2



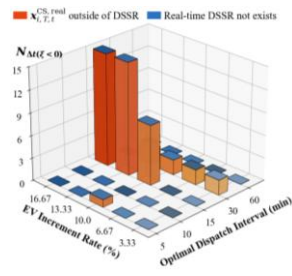
(b) EVCS3



(c) EVCS4



(d) EVCS5



(e) EVCS6

Fig. 13 M2 cumulative times of negative real-time security capacity for EVCS2 ~ EVCS6 (141-bus)